2.4.1 Prove Lemma 2.4.1: Modular curve $X(1) = SL_2(\mathbb{Z}) \backslash \mathcal{H}^*$ has one cusp.

$\forall \Gamma$ $X(\Gamma)$ has finitely many cusps.

Pf: For $\Gamma = SL_2(\mathbb{Z})$, only need $\forall \frac{r}{s} \in \mathbb{Q}$ $\gcd(r,s)=1$ $\exists \gamma \in \Gamma$ $\gamma \frac{r}{s} = \infty$

Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $\gamma \frac{r}{s} = \dfrac{a\frac{r}{s}+b}{c\frac{r}{s}+d} = \dfrac{ar+bs}{cr+ds}$

So $\begin{pmatrix} s-k & -(r+l) \\ s & -r \end{pmatrix} \in \Gamma$ works. it has det

$-sr+kr+sr+sl = kr+ls = 1$.

2.4.2 (a) Justify $\mathrm{Im}\,\gamma(\tau) \le \max\{\mathrm{Im}\,\tau, 1/\mathrm{Im}\,\tau\}$ $\tau \in \mathcal{H}$ $\gamma \in SL_2(\mathbb{Z})$.

used in Prp 2.4.2

Pf: Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$. $\mathrm{Im}\,\gamma(\tau) = \dfrac{\mathrm{Im}(\tau)}{|c\tau+d|^2}$.

$|\cancel{c\tau+d}|^2 = \dfrac{\mathrm{Im}\,\tau}{(c\,\mathrm{Re}\,\tau+d)^2 + (c\,\mathrm{Im}\,\tau)^2}$

If $|c| \ge 1$, then $\mathrm{Im}\,\gamma(\tau) \le \dfrac{\mathrm{Im}\,\tau}{(c\,\mathrm{Im}\,\tau)^2} = \dfrac{1}{c^2\,\mathrm{Im}\,\tau} \le \dfrac{1}{\mathrm{Im}\,\tau}$.

Otherwise, $c=0$, then $\mathrm{Im}\,\gamma(\tau) \le \dfrac{\mathrm{Im}\,\tau}{(c\,\mathrm{Re}\,\tau+d)^2} = \dfrac{\mathrm{Im}\,\tau}{d^2} \le \mathrm{Im}\,\tau$.
(so $a, d \ne 0$)

(b) Justify $\pi(U_1)$ and $\pi(U_2)$ are disjoint

Pf: Argue by contradiction. $\pi: \mathcal{H} \cup \mathbb{P}^1(\mathbb{Q}) \to \Gamma\backslash(\mathcal{H} \cup \mathbb{P}^1(\mathbb{Q}))$

$\tau \mapsto \Gamma\tau$ ~~$\cup\{\infty\}$~~.

$\frac{p}{q} \mapsto \frac{r}{s}$.

If $\pi(U_1) \cap \pi(U_2) \ne \phi$

then $\exists \tau_1 \in U_1$, $\tau_2 \in U_2$, $\gamma \in \Gamma$ $\gamma\tau_1 = \tau_2$. Since $U_1 = \alpha(N_M \cup \{\infty\})$

$\tau_1 = \alpha(\sigma)$, $\sigma \in N_M \cup \{\infty\}$.

Thus $\sigma = (\alpha^{-1}\gamma^{-1})\tau_2 \in SL_2(\mathbb{Z})\, U_2$.

However $SL_2(\mathbb{Z})K \cap N_M = \phi \Rightarrow SL_2(\mathbb{Z})\, U_2 \cap (N_M \cup \{\infty\}) = \phi$. gives contradiction

2.4.3 Show $D^* = D \cup \{\infty\}$ is compact in the $\mathcal{H}^*$ topology.

Pf: Given any cover $\{U_i\}$ of D^*,

Since ∞ is covered, the cover must has one open set of the form $N_M \cup \{\infty\}$.

Then proving D^* is compact is equivalent to proving $\tilde{D} := \left\{ \tau \in \mathcal{H} : \begin{array}{l} |\mathrm{Re}\,\tau| \le \frac{1}{2}, \ |\tau| \ge 1, \\ \mathrm{Im}\,\tau \le M \end{array} \right\}$

is compact.

A closed bounded set is compact in $\mathbb{C}$, hence in $\mathcal{H}$.

2.4.4 (a) Show the width h_s is finite.

Pf: $h_s := \left| SL_2(\mathbb{Z})_\infty / (\delta\{\pm I\}\Gamma\delta^{-1})_\infty \right|$.

Since $SL_2(\mathbb{Z})_\infty = \{\pm I\}\left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle \simeq \mathbb{Z}$ ~~for~~

$(\delta\{\pm I\}\Gamma\delta^{-1})_\infty = \{\pm I\}(\delta\Gamma\delta^{-1})_\infty = \{\pm I\}\left\langle \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix} \right\rangle$.

for some $h > 0$, is a sub group of $SL_2(\mathbb{Z})_\infty$, hence $\simeq h\mathbb{Z}$.

Thus $SL_2(\mathbb{Z})_\infty / (\delta\{\pm I\}\Gamma\delta^{-1})_\infty \simeq \mathbb{Z}/h\mathbb{Z}$. finite.

(b) Show $h_s = \left| SL_2(\mathbb{Z})_s / \{\pm I\}\Gamma_s \right|$. Recall $\delta s = \infty$

Pf: $(\delta\{\pm I\}\Gamma\delta^{-1})_\infty = \{\pm I\}(\delta\Gamma\delta^{-1})_\infty = \{\pm I\}\Gamma_s$.

$SL_2(\mathbb{Z})_\infty = (\delta SL_2(\mathbb{Z})\delta^{-1})_s = SL_2(\mathbb{Z})_s$.

(c) Show if $s \in \mathbb{Q} \cup \{\infty\}$ and $\gamma \in SL_2(\mathbb{Z})$ then $h_{\gamma(s)}$ under $\gamma\Gamma\gamma^{-1}$ is the same as h_s under Γ

$\gamma : s \mapsto \gamma s$

Pf: $h_{\gamma(s), \gamma\Gamma\gamma^{-1}} = \left| SL_2(\mathbb{Z})_\infty / \{\pm I\}(\delta_{\gamma(s)}\gamma\Gamma\gamma^{-1}\delta_{\gamma(s)}^{-1})_\infty \right|$. $\delta_{\gamma(s)} : \gamma(s) \mapsto \infty$.

Thus $\delta_{\gamma(s)}\gamma : s \mapsto \infty$. $h_{\gamma(s), \gamma\Gamma\gamma^{-1}} = \left| SL_2(\mathbb{Z})_\infty / \{\pm I\}(\delta_s\Gamma\delta_s^{-1})_\infty \right|$.

$\Rightarrow \delta_{\gamma(s)}\gamma = \delta_s$

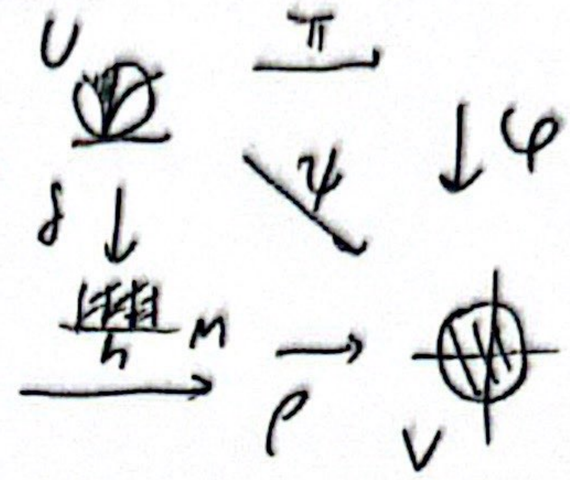
Another method: by (b), $\gamma\Gamma_{\gamma(s)}\gamma^{-1} = \gamma\{\gamma' : \gamma'\gamma(s) = \gamma(s)\}\gamma^{-1} = \{\gamma' : \gamma'(s) = s\}$

$= \Gamma_s$.

2.4.5 Explain $\varphi : \pi(U) \to V$ as defined is a homeomorphism.

Pf: $\psi = \varphi \circ \pi$. π open continuous.

ψ open continuous.



φ injective, surjective.

2.4.6 Show Λ_2 contains no elliptic points for $SL_2(\mathbb{Z})$.

(in Hint: for $\tau = i$ or μ_3, $|c\tau+d| \geq 1$ $\forall$ $(c,d)\neq(0,0)$).

Pf: $\operatorname{Im}\gamma\tau = \dfrac{\operatorname{Im}\tau}{|c\tau+d|^2}$ $\exists\, \tau$ elliptic $\Rightarrow \exists\, \gamma$ $\gamma\tau = \tau$.

$\Rightarrow \operatorname{Im}\gamma\tau = \operatorname{Im}\tau$

$\Rightarrow 1 = |c\tau+d|^2 = (c\operatorname{Re}\tau + d)^2 + (c\operatorname{Im}\tau)^2 \overset{c\neq 0}{\geq} (\operatorname{Im}\tau)^2 > 2^2$ Contradiction

$\overset{c=0}{=} d^2 \Rightarrow \gamma = \pm I$.

(So actually Λ_1 works)

For $\tau = i$ or μ_3, $|c\tau+d|^2 = (c\operatorname{Re}\tau+d)^2 + (c\operatorname{Im}\tau)^2$

$\overset{\tau=i}{=} c^2$ $\quad \gamma i = i \Rightarrow \gamma \in \left\langle \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\rangle \Rightarrow c^2 = 1$.

$\overset{\tau=\mu_3}{=} \left(\frac{1}{2}c+d\right)^2 + \left(\frac{\sqrt{3}}{2}c\right)^2 = c^2+cd+d^2 > 0$.

$\gamma\mu_3 = \mu_3 \Rightarrow \gamma \in \left\langle \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \right\rangle$

$= \left\{ \pm\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \pm\begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \pm I \right\} \Rightarrow c^2+cd+d^2 = 3$ or 1.

2.4.7 $\gamma\Gamma_1\gamma^{-1} \subset \Gamma_2$, $\gamma \in GL_2^+(\mathbb{Q})$. Show $f : X(\Gamma_1) \to X(\Gamma_2)$, $\Gamma_1\tau \mapsto \Gamma_2\gamma(\tau)$ is holomorphic.

Pf: well defined: $\Gamma_1\tau = \Gamma_1\tau' \overset{g(\tau)=\tau'}{\Rightarrow} \gamma(\tau') = \gamma g(\tau) = \gamma g \gamma^{-1}(\gamma(\tau))$

with $\gamma g \gamma^{-1} \in \Gamma_2$. Thus $\Gamma_2\gamma(\tau') = \Gamma_2\gamma(\tau)$.

maps cusps to cusps: $\gamma \in GL_2^+(\mathbb{Q}) \Rightarrow \gamma(\mathbb{P}^1(\mathbb{Q})) = \mathbb{P}^1(\mathbb{Q})$.

Continuity: $\tilde{f}(\tau) = \gamma(\tau) = \dfrac{a\tau+b}{c\tau+d}$ holomorphic.

$\tilde{f}$ is Γ_1-equivariant: $\tilde{f}(g\tau) = \gamma g(\tau) = (\gamma g \gamma^{-1})\tilde{f}(\tau)$ $\quad g \in \Gamma_1$

$\gamma g \gamma^{-1} \in \Gamma_2$.

(2.4.7) Thus $\tilde{f}$ descends to a continuous map f.

f holomorphic: Let $[\tau_0] \in X(\Gamma_1)$. pick $U \ni \tau_0$.

such that U simply connected

and stabilizer $\Gamma_{1,\tau_0} U = U$ acts trivially.

Then the quotient $\pi: U \to \Gamma_1 \backslash U$ is biholomorphic

On such neighborhood, $f = \pi_2 \circ \tilde{f} \circ \pi_1^{-1}$

Since $\tilde{f}$ holomorphic ~~as two~~ as well, f is holomorphic

f holomorphic at cusp: near $s \in \mathbb{P}^1(\mathbb{Q})$, a cusp,

choose $\delta = \delta_s : s \mapsto \infty$.

under $\tau \mapsto \gamma(\tau)$, $q = e^{2\pi i \tau / h_s}$ (local coordinates) $\mapsto q' = e^{2\pi i \gamma(\tau)/h_{s'}}$

holomorphic in q. Hence f holomorphic at cusps.